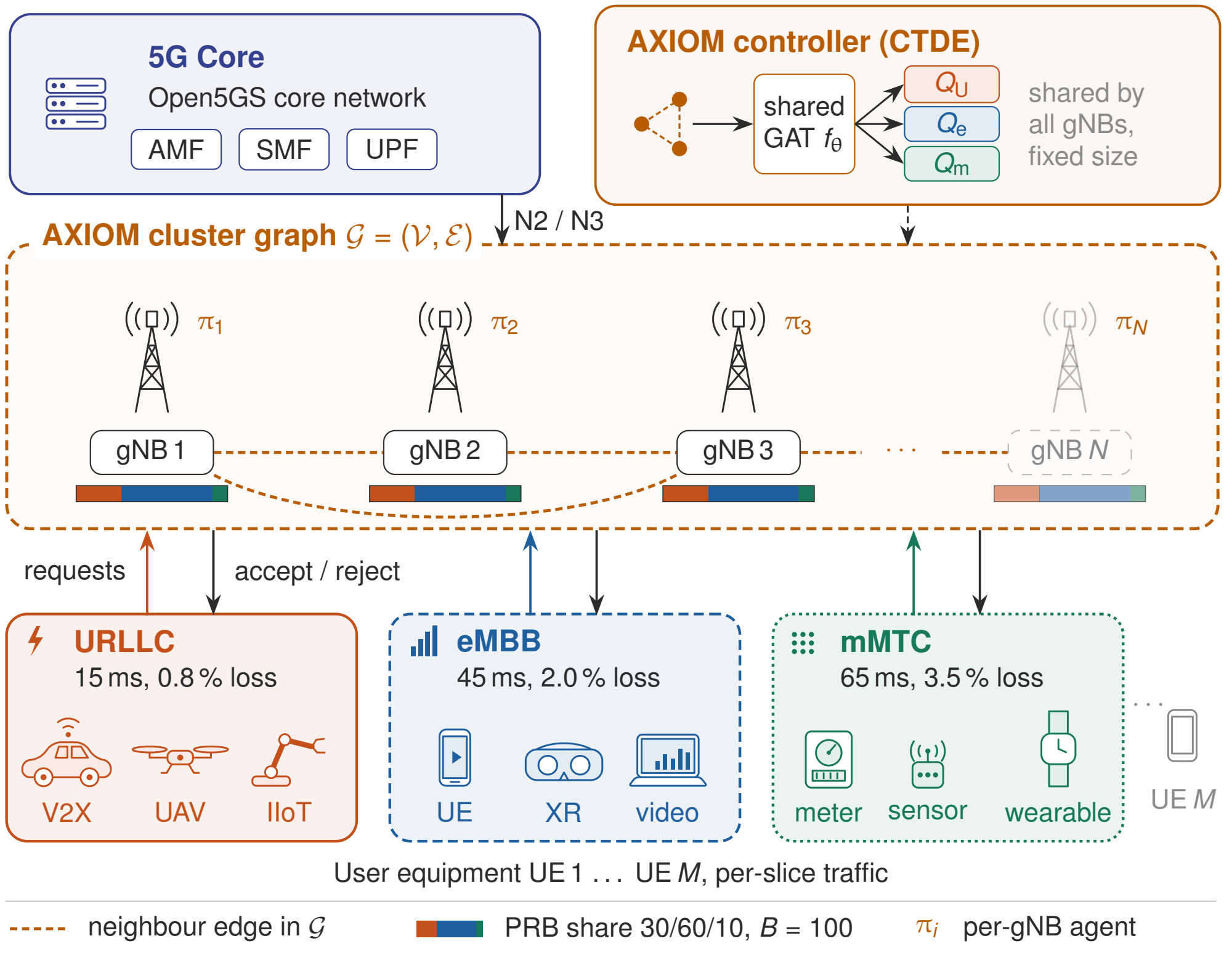

5G Core
Open5GS core network
AMF
SMF
UPF
AXIOM controller (CTDE)
shared GAT f_θ
Q_U
Q_e
Q_m
shared by all gNBs, fixed size
N2 / N3
AXIOM cluster graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
π_1
π_2
π_3
π_N
gNB 1
gNB 2
gNB 3
gNB N
requests
accept / reject
URLLC
15 ms, 0.8 % loss
V2X
UAV
IIoT
eMBB
45 ms, 2.0 % loss
UE
XR
video
mMTC
65 ms, 3.5 % loss
meter
sensor
wearable
UE M
User equipment UE 1 . . . UE M, per-slice traffic
neighbour edge in $\mathcal{G}$
PRB share 30/60/10, B = 100
π_i per-gNB agent